

Backtesting an ATM Option Straddle Rolling Strategy

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1 Strategy and Backtesting Framework

For each trading day, the backtest reads the near-month futures file (NIFTY-I.csv or BANKNIFTY-I.csv) and identifies the option expiry closest to that date. At every available one-second timestamp, the futures price is used to select the closest available strike price.

The strategy buys one Call Option (CE) and one Put Option (PE) at the selected at-the-money (ATM) strike. When the nearest ATM strike changes, the existing CE and PE are sold at their latest available prices and a new CE-PE pair is bought at the new strike. All open positions are closed at the end of each day.

- **Position size:** Maximum one position per option instrument.
- **Instruments:** NIFTY and BANKNIFTY nearest-expiry options.
- **PnL:** Total PnL = Realized PnL + Unrealized Mark-to-Market PnL.
- **Roll:** Closing the old ATM straddle and opening the newly selected ATM straddle.

2 Portfolio-Level Performance

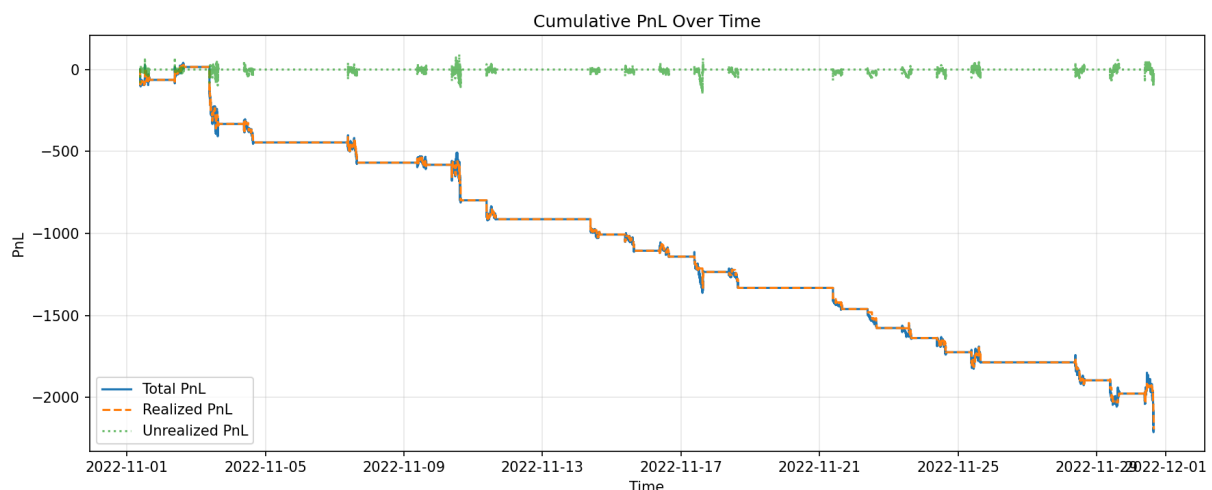


Figure 1: Cumulative total, realized, and unrealized PnL over the month.

The cumulative PnL curve declines from near zero to approximately **-2200 points** by month-end. The total PnL and realized PnL almost overlap, which is expected because positions are repeatedly closed during strike rolls and fully squared off at the end of each day. Unrealized PnL remains close to zero most of the time because open positions are short-lived relative to the number of roll events.

The steadily downward shape indicates that losses were not caused by one isolated event; instead, the strategy experienced repeated small losses across the month. This suggests that frequent ATM strike switching was costly for this long-straddle strategy under the observed market conditions.

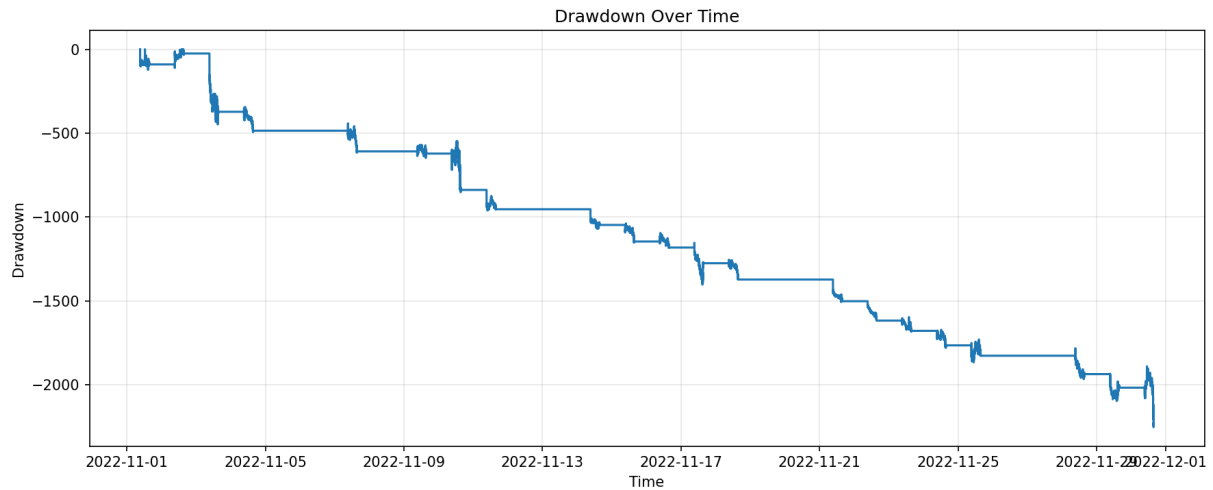


Figure 2: Drawdown of cumulative PnL from its running peak.

Drawdown measures the loss from the best cumulative PnL achieved so far. The maximum drawdown reaches roughly **-2200 points**. Throughout the month, the strategy was unable to recover the losses and had a steady drop as the losses piled up.

3 Daily and Underlier-Level Analysis

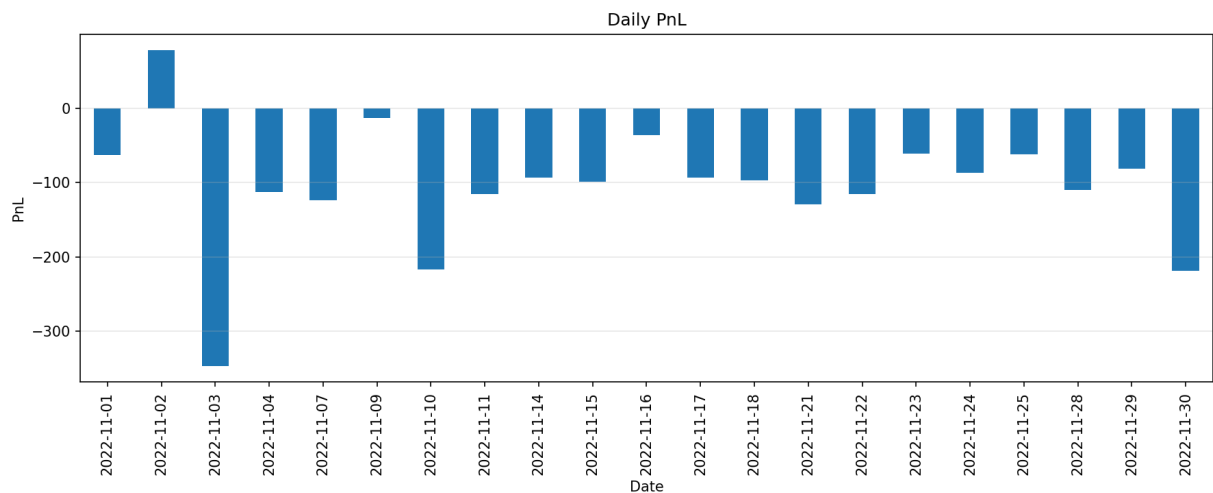


Figure 3: Daily realized PnL across all trading days.

The daily PnL chart shows only one clearly positive day, while nearly every other day is negative. The largest daily loss occurs around November 3, at approximately **-350 points**. Almost all the days have losses between 50 to 130 points.

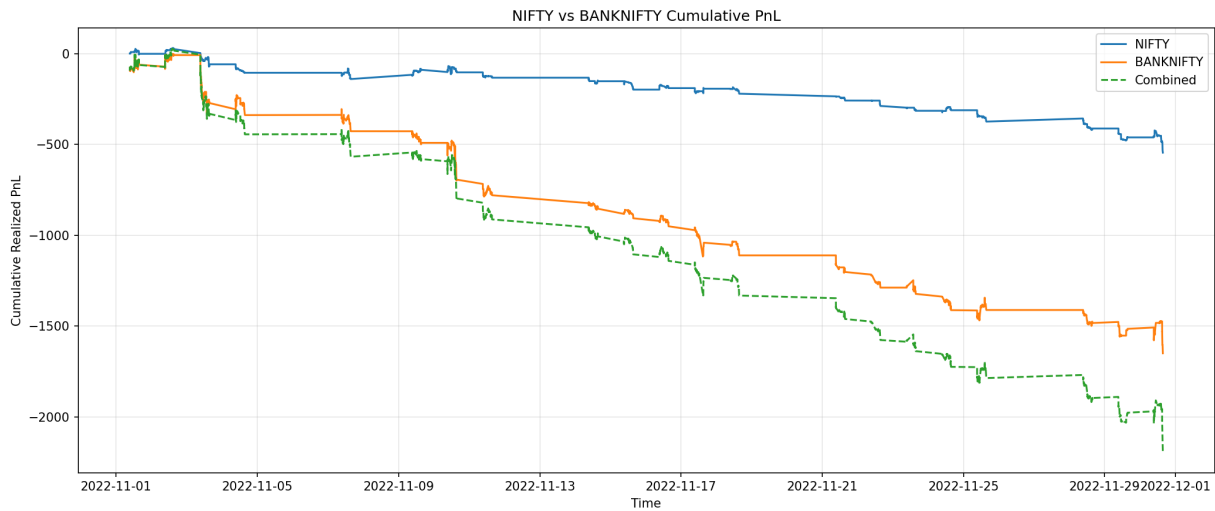
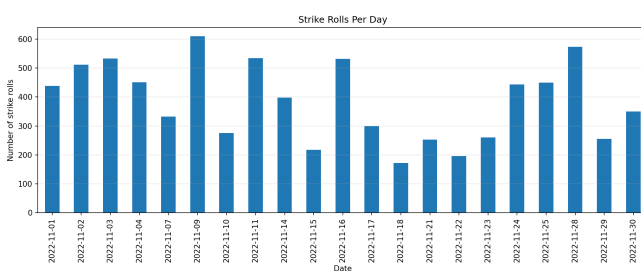


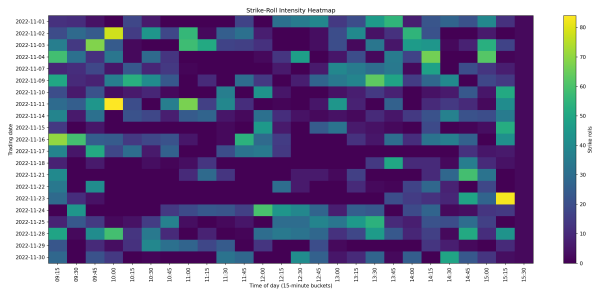
Figure 4: Cumulative realized PnL contribution from NIFTY, BANKNIFTY, and the combined portfolio.

Both underliers contribute negatively, but **BANKNIFTY is the dominant source of losses**, ending near -1600 points, while NIFTY ends near -550 points. Thus, despite having the same strategy, BANKNIFTY performed significantly worse compared to NIFTY.

4 Trading Activity and Intraday Behaviour



(a) Strike rolls per day



(b) Intraday strike-roll intensity

Figure 5: Trading activity generated by ATM strike changes.

The strategy generates a large number of strike rolls, ranging from approximately 170 to more than 600 rolls per day. Since every roll closes one CE and one PE, then opens a new CE and PE, this represents very high turnover.

The strike-roll heatmap shows that roll activity is distributed throughout the trading session rather than being limited to market open or close.

5 Conclusion

The backtest implements the required nearest-expiry ATM CE–PE rolling strategy for NIFTY and BANKNIFTY. The results show a cumulative loss, large unrecovered drawdown, and consistently negative daily PnL. BANKNIFTY accounts for most of the portfolio loss, while the high number of strike rolls indicates excessive turnover.

A better strategy would be to introduce a **roll buffer**: instead of changing strike immediately whenever the nearest strike changes, the strategy could roll only after the futures price moves a specified distance beyond the current strike boundary.